Unit 1

1. Introduction to AI

Module 1.9

1.9 Standards, Regulations and AI

Tutorial 1.9

1.9 Standards, Regulations and AI

AI-1.9.1 K2 Describe how standards apply to AI-based systems.

Why Standards and Regulations matter in AI

- Standards ensure safety, reliability, and ethical behavior in AI applications
- Standards and regulations also promote transparency, accountability, and trustworthiness in AI
- It supports compliance with legal norms
- It enable consistent testing practices across diverse AI systems
- It also helps to mitigate risks like bias, discrimination, and unintended behavior

Standards, Regulations and AI

- The Joint Technical Committee of IEC and ISO on information technology (ISO/IEC JTC1) prepares international standards which contribute towards AI.
- A subcommittee on AI (ISO/IEC JTC 1/SC42), was set up in 2017.
- ISO/IEC JTC1/SC7, which covers software and system engineering, has published a technical report on the “Testing of AI-based systems”
- The EU-wide General Data Protection Regulation (GDPR) came into effect in May 2018 and sets obligations for data controllers with regards to personal data and automated decision-making.
- GDPR includes requirements to assess and improve AI system functional performance, including the mitigation of potential discrimination, and for ensuring individuals rights are not subjected to automated decision-making.
- The most important aspect of the GDPR from a testing perspective is that personal data (including predictions) should be accurate.

Standards, Regulations and AI Contd.

- The German national standards body (DIN) has also developed the AI Quality Metamodel
- Standards on AI are also published by industry bodies.
  - The Institute of Electrical and Electronics Engineers (IEEE) is working on a range of standards on ethics and AI (The IEEE Global Initiative for Ethical Considerations in Artificial Intelligence and Autonomous Systems.
- Where AI is used in safety-related systems, the relevant regulatory standards are applicable, such as ISO 26262 and ISO/PAS 21448 (SOTIF) for automotive systems.
- Such regulatory standards are typically mandated by government bodies, and it would be illegal to sell a car in some countries if the included software did not comply with ISO 26262.

THANK YOU!!!